Radar Signal Classification Using Machine Learning and Neural Networks

Project Summary & Performance Report

Made by: Sitah Alyahya

Supervised by: Raghad Alsammari

1. Introduction

This project focuses on the automated classification of radar signals using machine learning models and artificial neural networks. The core objective was to accurately classify detected radar targets into three distinct operational categories: Drones, Cars, and People. By analyzing unique radar reflection characteristics, this project explores how artificial intelligence can significantly enhance target detection capability, reduce false alarm rates, and improve the reliability of modern automated target detection systems.

2. Dataset Description

The dataset used in this study consists of real-world radar signal measurements captured via a Frequency-Modulated Continuous-Wave (FMCW) radar system. Each observation in the dataset represents a processed radar signal signature transformed into descriptive numerical attributes.

After aggregating the raw source data files, the final combined dataset comprised approximately 192,335 samples across 62 total columns, including 61 distinct numerical features and 1 target class label column. The dataset distribution reveals a well-balanced distribution across the three target classes, minimizing classification bias. No missing or null values were identified during data cleaning.

- Drones: 55,715 samples
- Cars: 62,920 samples
- People: 73,700 samples

3. Models Used

Four distinct traditional machine learning algorithms were trained and rigorously evaluated to establish solid baselines for radar target classification. Performance was measured using four standardized metrics: Accuracy, Precision, Recall, and F1-Score.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.566	0.565	0.566	0.564
Random Forest	0.744	0.771	0.744	0.748
KNN	0.760	0.781	0.760	0.766
Decision Tree	0.743	0.771	0.743	0.742

Model Comparison Analysis: Logistic Regression yielded the poorest results because radar target backscatter patterns are fundamentally complex and non-linearly separable. Tree-based ensemble methods (Random Forest) and non-parametric algorithms (K-Nearest Neighbors) successfully captured non-linear feature interactions, achieving significantly higher accuracy. Out of all traditional baselines, KNN demonstrated superior classification capacity with the fewest prediction errors.

4. Neural Network

To further enhance target identification performance, a Deep Neural Network (DNN) was developed utilizing TensorFlow and Keras. The network was engineered with multiple dense, fully-connected layers designed to automatically discover hierarchical feature representations from raw scaled radar inputs.

The architectural layout and training specifics include:
• Layer 1: Dense Input Layer (128 neurons, ReLU activation function)
• Layer 2: Dropout Layer (Rate of 0.3 to mitigate overfitting by random neuron deactivation)
• Layer 3: Dense Intermediate Layer (64 neurons, ReLU activation function)
• Layer 4: Dense Output Layer (3 neurons corresponding to the target classes, Softmax activation)

Prior to model feeding, features were standardized using StandardScaler to achieve uniform variance. To optimize training efficiency and avoid over-training, an EarlyStopping callback was incorporated to continuously monitor the validation loss, halting the training process immediately when performance plateaued.

Metric	Neural Network Value
Accuracy	0.788
Precision	0.802
Recall	0.788
F1-Score	0.790

The resulting neural network training and validation curves remained stable, showing smooth convergence without evidence of overfitting or variance issues.

5. Results & Comparison

An exhaustive comparative analysis reveals that the Neural Network configuration achieved the highest classification performance across all metrics, noticeably outperforming all traditional machine learning methodologies.

Final Performance Ranking:
1. Neural Network (Best overall capability)
2. K-Nearest Neighbors (KNN) | 3. Random Forest | 4. Decision Tree | 5. Logistic Regression

The Deep Neural Network succeeded where other models struggled because its interconnected structure can map non-linear relationships and high-dimensional interactions within the 61 FMCW radar features much more efficiently than traditional parametric or baseline models.

6. What I Learned

This project provided critical insights into the practical end-to-end data science and engineering workflow for high-frequency signal datasets:

- Efficiently consolidating, balancing, and cleaning large multi-file datasets containing hundreds of thousands of entries.
- The crucial impact of proper data preprocessing, establishing that feature scaling via standardizers drastically improves convergence rates.
- Designing, tuning, and training multi-layer deep learning networks using TensorFlow and Keras framework architectures.
- Applying robust regularization techniques like Dropout layers and EarlyStopping to confidently avoid overfitting.
- Conducting holistic model evaluations utilizing comprehensive metric reports including Accuracy, Precision, Recall, and F1-Scores.